

Run 10 | TEA

India

IV + controls

Target Total early-stage entrepreneurial activity; binary 1 vs. 0

Sample 27,555 respondents **Coverage** 1 country; 2015–2022

Positive cases 3,188 (11.57%) **Case time** 52s

Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.9974). Highest F1: Random Forest (0.9419). For H2O GBM, adding controls changed ROC-AUC by +0.0009 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.9974	0.9777	98.64	93.36	94.98	0.9416	3s
Random Forest	0.9974	0.9788	98.64	92.97	95.45	0.9419	1s
XGBoost	0.9974	0.9774	98.49	91.34	96.08	0.9365	<1s
Logistic Regression	0.9973	0.9771	98.55	92.78	94.82	0.9379	1s
CatBoost	0.9973	0.9763	98.62	92.69	95.60	0.9413	6s
Ridge	0.9972	0.9751	98.60	92.17	96.08	0.9408	11s
Elastic Net	0.9971	0.9751	98.48	92.09	94.98	0.9351	11s
Lasso	0.9971	0.9747	98.42	91.79	94.82	0.9328	12s
Decision Tree	0.9940	0.9196	98.48	91.58	95.60	0.9355	<1s
Naive Bayes	0.9380	0.6795	92.81	64.86	82.57	0.7265	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,739.61	54.18
2	Established business owner	621.53	19.36
3	Occupational status	117.62	3.66
4	Age	61.86	1.93
5	Business angel	61.41	1.91
6	Knowing entrepreneurs	58.13	1.81
7	Business exit	45.40	1.41
8	Household size	40.33	1.26

Share is the percentage of total forest impurity importance across all 40 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 10_obj1_india_iv_cv_TEA. Scores and importance are taken directly from saved CSV outputs.